

Mini Project – Breast Cancer Classification

Artificial Intelligence Internship | Machine Learning Model Comparison

1. Problem Statement

Breast cancer is a major health concern, and accurate classification of tumors can assist in identifying whether a tumor is malignant or benign. The objective of this project is to use Machine Learning classification algorithms to predict the type of breast tumor from numerical diagnostic measurements.

2. Dataset Description

The Breast Cancer Wisconsin Diagnostic dataset available through Scikit-learn was used. It contains 569 observations and 30 numerical features describing characteristics of cell nuclei, including radius, texture, perimeter, area, smoothness, and other measurements. The target variable contains two classes: malignant and benign.

3. Data Preprocessing

The dataset was examined for missing values and duplicate records. No missing values were found. All 30 numerical diagnostic features were selected as input variables. The dataset was divided into 80% training data and 20% testing data using stratified sampling and a fixed random state to make the experiment reproducible.

4. Models Selected

Logistic Regression: Selected as a baseline classification algorithm for binary classification. It estimates class probabilities and is computationally efficient.

Decision Tree Classifier: Selected as a second model because it can capture non-linear relationships and provides an interpretable tree structure and feature importance values.

5. Evaluation Metrics

The models were evaluated using Accuracy, Precision, Recall, and Confusion Matrix. Accuracy measures overall correctness, Precision measures the proportion of positive predictions that are correct, and Recall measures how effectively the model identifies the positive class.

6. Results

Model	Accuracy	Precision	Recall
Logistic Regression	96.49%	95.95%	98.61%
Decision Tree	92.11%	95.65%	91.67%

Best-performing model: Logistic Regression. It achieved the highest accuracy and recall among the two models. Its accuracy was 96.49%, compared with 92.11% for the Decision Tree. Logistic Regression also achieved a higher recall of 98.61%.

7. Analysis and Comparison

Logistic Regression performed better overall on the test dataset. The difference in accuracy was approximately 4.38 percentage points. Logistic Regression also had a higher recall, meaning it identified a larger proportion of the relevant test samples. The Decision Tree had a slightly lower precision than Logistic Regression and a noticeably lower recall. The Decision Tree is nevertheless useful because its decisions are easier to represent as rules and it provides feature importance information.

8. Visualizations Included in the Project

- Target class distribution
- Logistic Regression confusion matrix
- Decision Tree confusion matrix
- Model performance comparison
- Top 10 Decision Tree feature importances

9. Conclusion

Two Machine Learning classification models were developed to classify breast tumors as malignant or benign. Based on the test-set evaluation, Logistic Regression was the best-performing model, achieving 96.49% accuracy, 95.95% precision, and 98.61% recall. Therefore, Logistic Regression is selected as the preferred model for this dataset.

10. Limitations and Future Improvements

The results are based on a single dataset and one train-test split, so they should not be interpreted as evidence of clinical effectiveness. The model should not replace professional medical diagnosis. Future work could include cross-validation, hyperparameter tuning, feature scaling and selection, testing additional algorithms, and validating the models on independent clinical datasets.

Dataset Source

Breast Cancer Wisconsin Diagnostic Dataset – provided through Scikit-learn.